HANYUN HU | UCL

Transport accessibility and robbery in London

Predicting area-level rates beyond baseline neighbourhood characteristics

THE QUESTION

Do public transport accessibility and proximity to rail-based stations improve the prediction of robbery rates across London LSOAs?

LSOA = Lower-layer Super Output Area, a small-area statistical geography.

STUDY DESIGN

- Predictive comparison of baseline and transport-extended models.
- Multi-source data integration, geographical coverage checks and nearest-station distance calculation.
- Cross-model evaluation, model interpretation and spatial sensitivity analysis.

4,988

London LSOAs in the common sample

2023

Crime counts and residential population

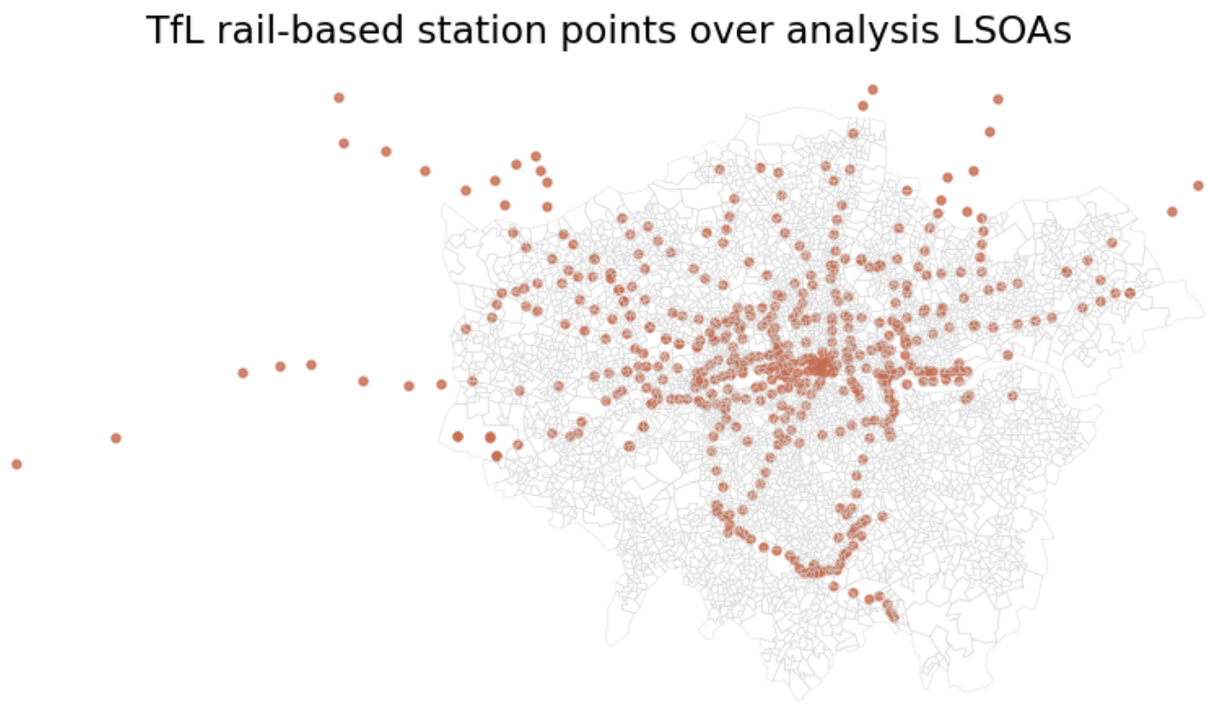
3

Model families compared

Main finding: transport-extended models improved prediction across all three tested model families. This is predictive evidence, not a causal claim.

1 | Data integration and spatial coverage

Analysis sample defined by matched geographical identifiers and verified data coverage.



TfL rail-based station locations over analysis LSOA boundaries.

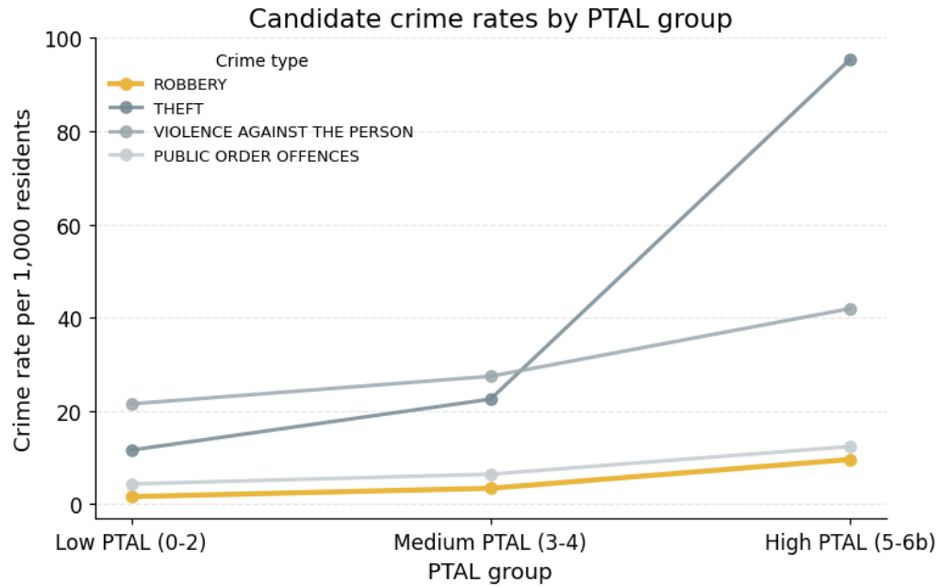
DATA AND DEFINITIONS

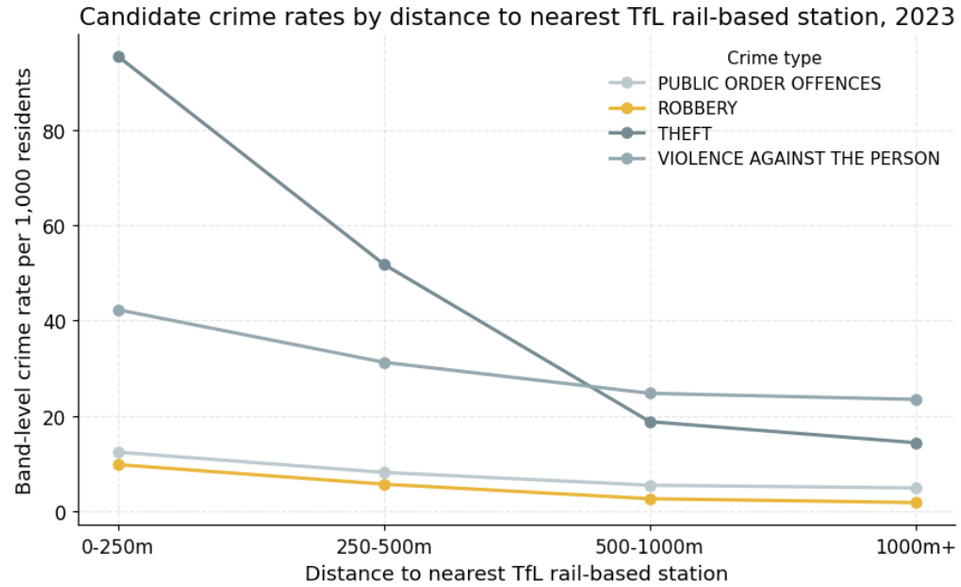
- MPS crime counts; 2023 population and age structure; IMD; TfL accessibility and rail-station locations.
- Common sample: 4,988 LSOAs. Six PTAL-covered areas had no corresponding crime-table record; missing was not assumed to mean zero.
- Projected geometry: EPSG:27700. Station proximity: straight-line distance from each LSOA centroid to the nearest TfL rail-based station.

Mean PTAL Accessibility Index measures public transport accessibility; centroid-to-station distance measures proximity. Neither directly measures passenger flows.

2 | Outcome selection and sensitivity analysis

Outcome screening examined central-area extremes and the suitability of resident population as the rate denominator.





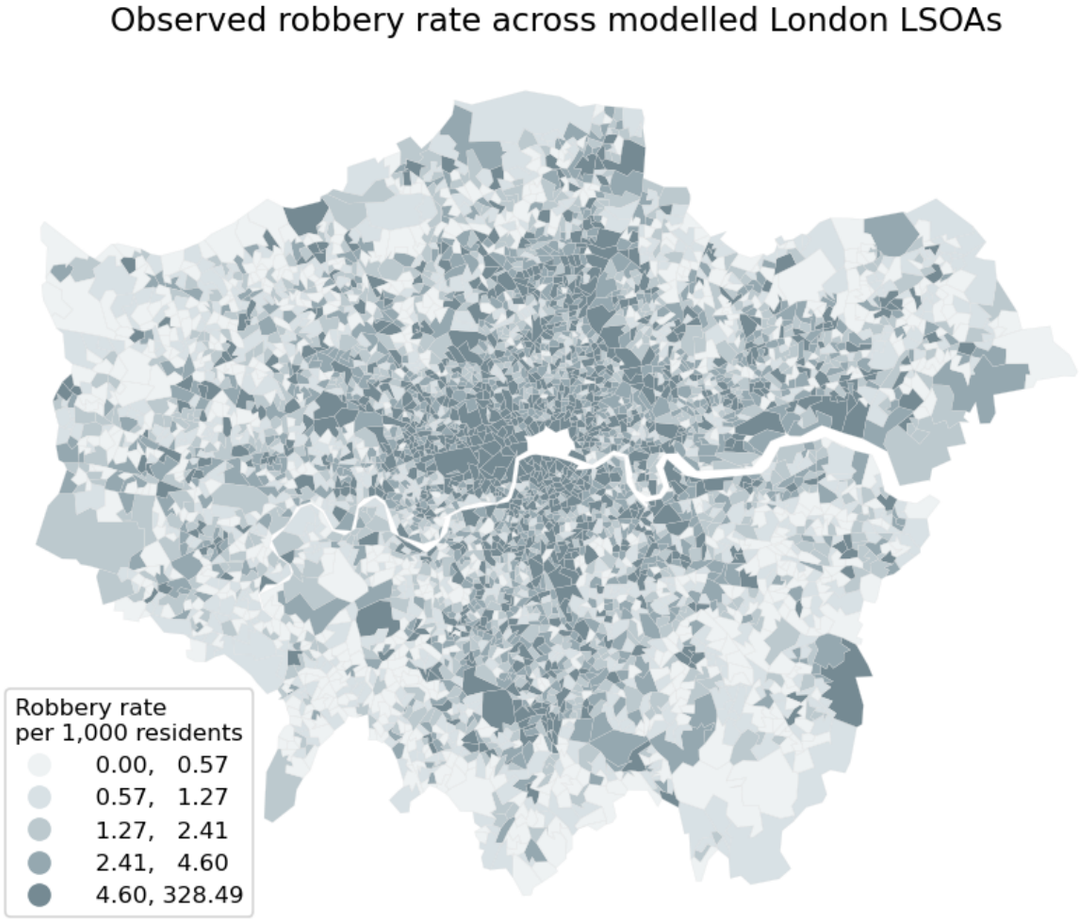
Candidate offences: robbery, theft, violence against the person and public order offences. Rates use residents as the denominator.

Robbery rate / 1,000 residents	0-250m	250-500m	500-1,000m	1,000m+
Original common sample	9.86	5.77	2.73	1.93
Excluding Westminster	8.01	4.97	2.67	1.93

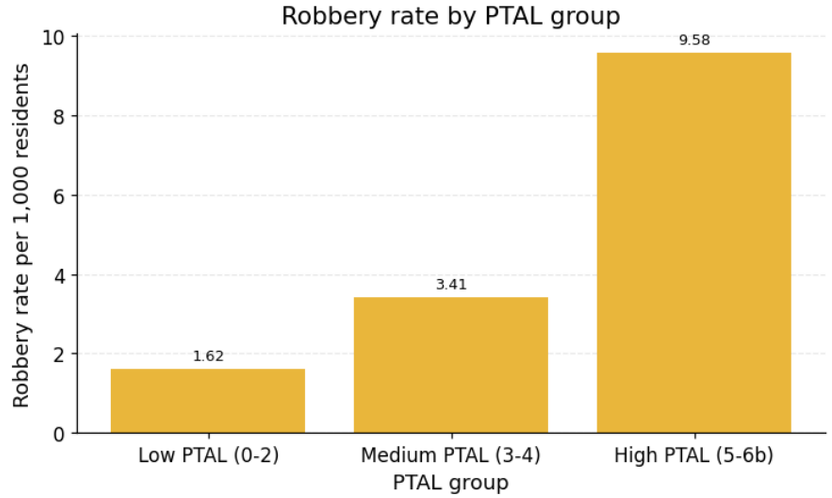
Robbery was selected using the literature, distributions and sensitivity results. Its station-distance gradient remained after excluding Westminster; resident-population exposure limitations remained.

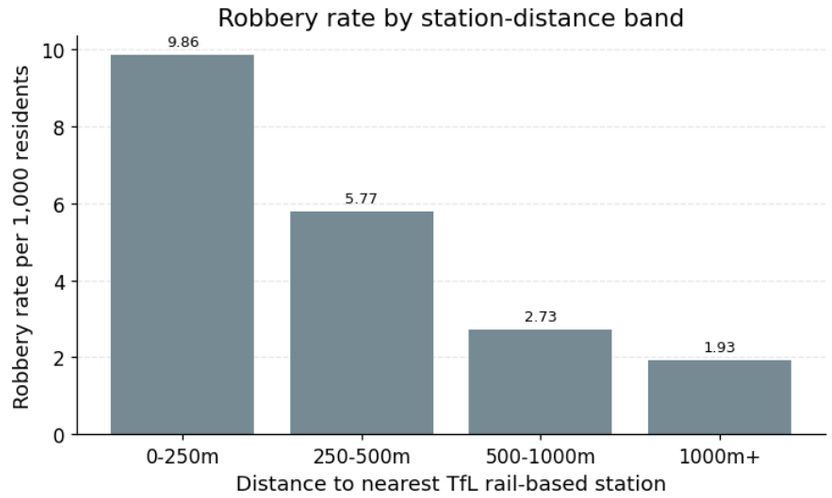
3 | Observed spatial patterns

Spatial and grouped robbery-rate patterns indicate associations, without establishing their causes.



Map: five quantile classes of observed LSOA robbery rates. Classes contain roughly equal numbers of areas, not equal rate intervals.

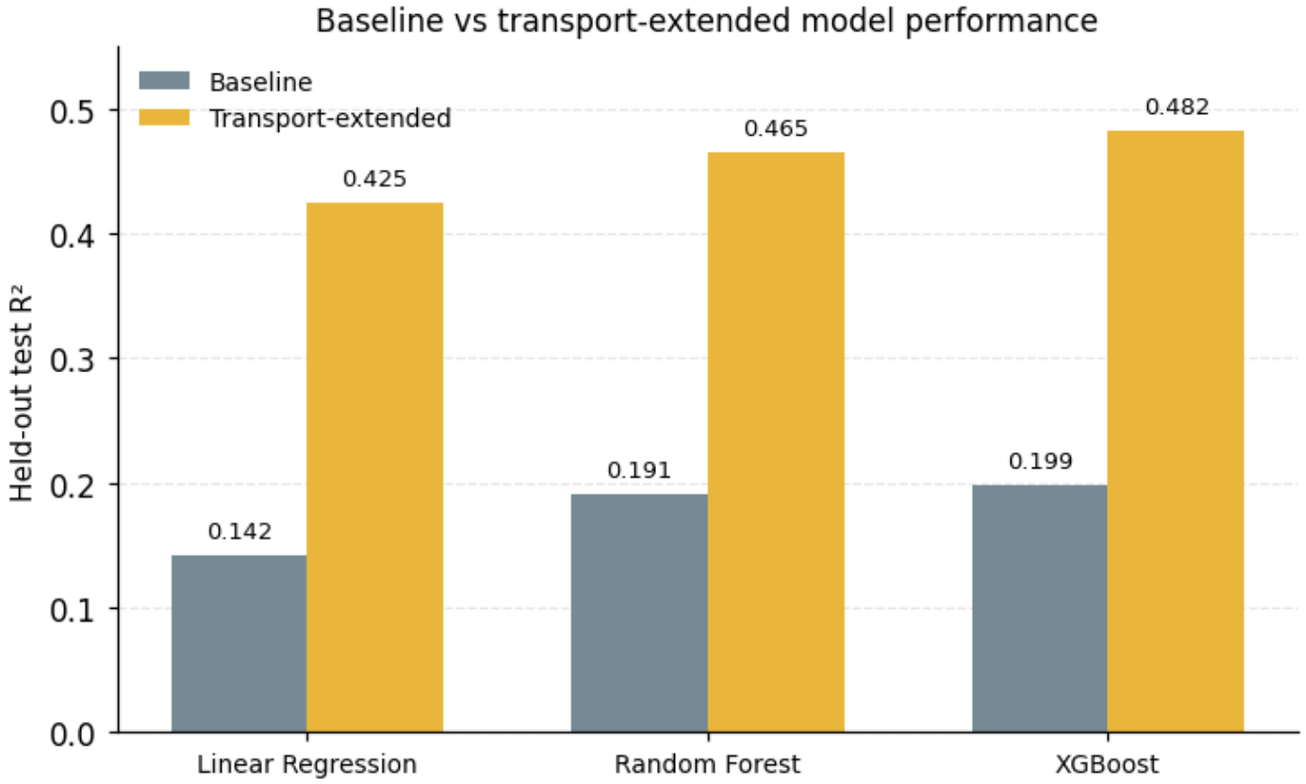




Bars: total offences / total residents in each group x 1,000; not the average of individual LSOA rates.

4 | Predictive contribution of transport indicators

Baseline and extended specifications compared within each model family under the fixed settings.



FEATURE SETS

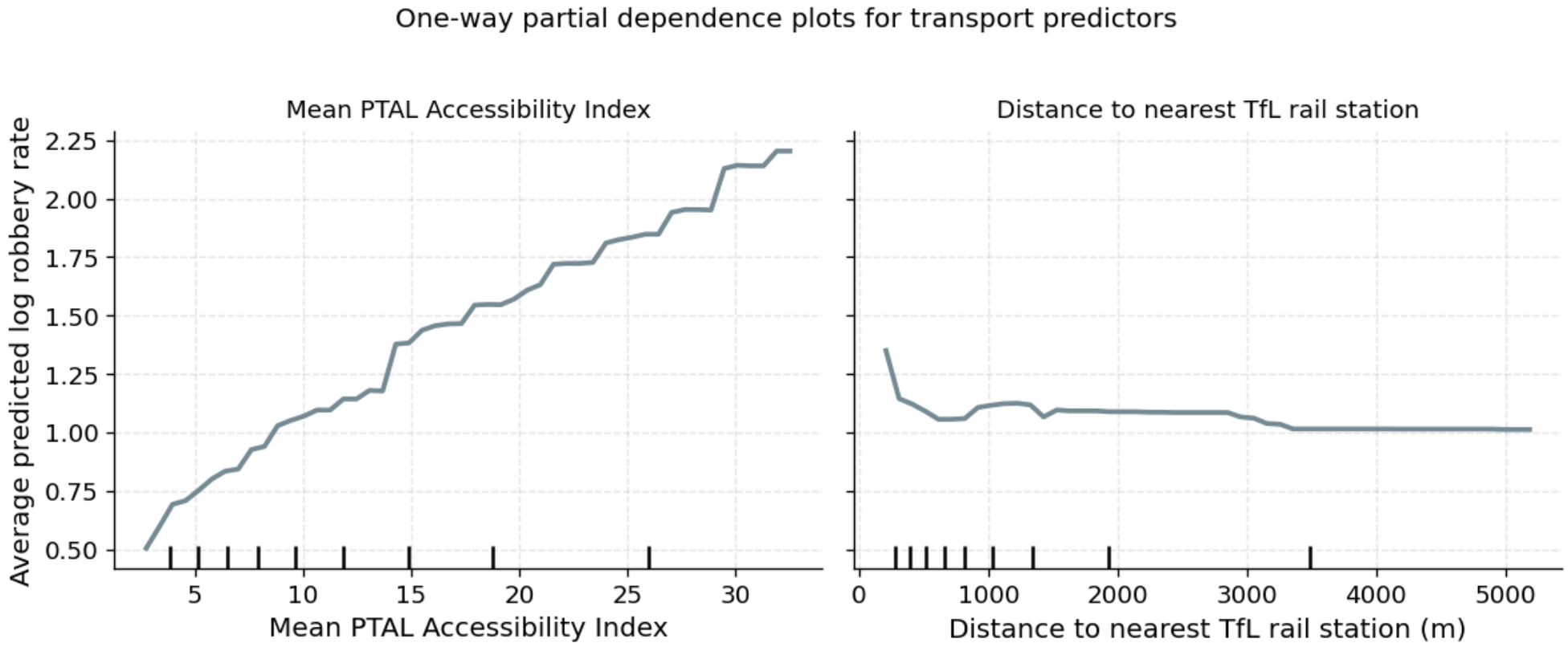
- **Baseline:** IMD score, log population density, share aged 16-24.
- **Transport-extended:** baseline + mean PTAL AI + nearest-station distance.
- **Model families:** Linear Regression, Random Forest and XGBoost.
- **Evaluation:** 75/25 train/test split; five-fold CV within training data.

Held-out test R-squared on log1p(robbery rate). Fixed-model comparison; not the tuned-model result.

Result: transport-extended specifications improved held-out prediction in every tested family. In the fixed XGBoost comparison, test R^2 rose from 0.199 to 0.482.

5 | Model interpretation

Final interpretation uses the tuned transport-extended XGBoost model.



PREDICTOR IMPORTANCE

Mean PTAL AI had the highest permutation importance among included features. Population density and deprivation also contributed.

MODEL RESPONSE

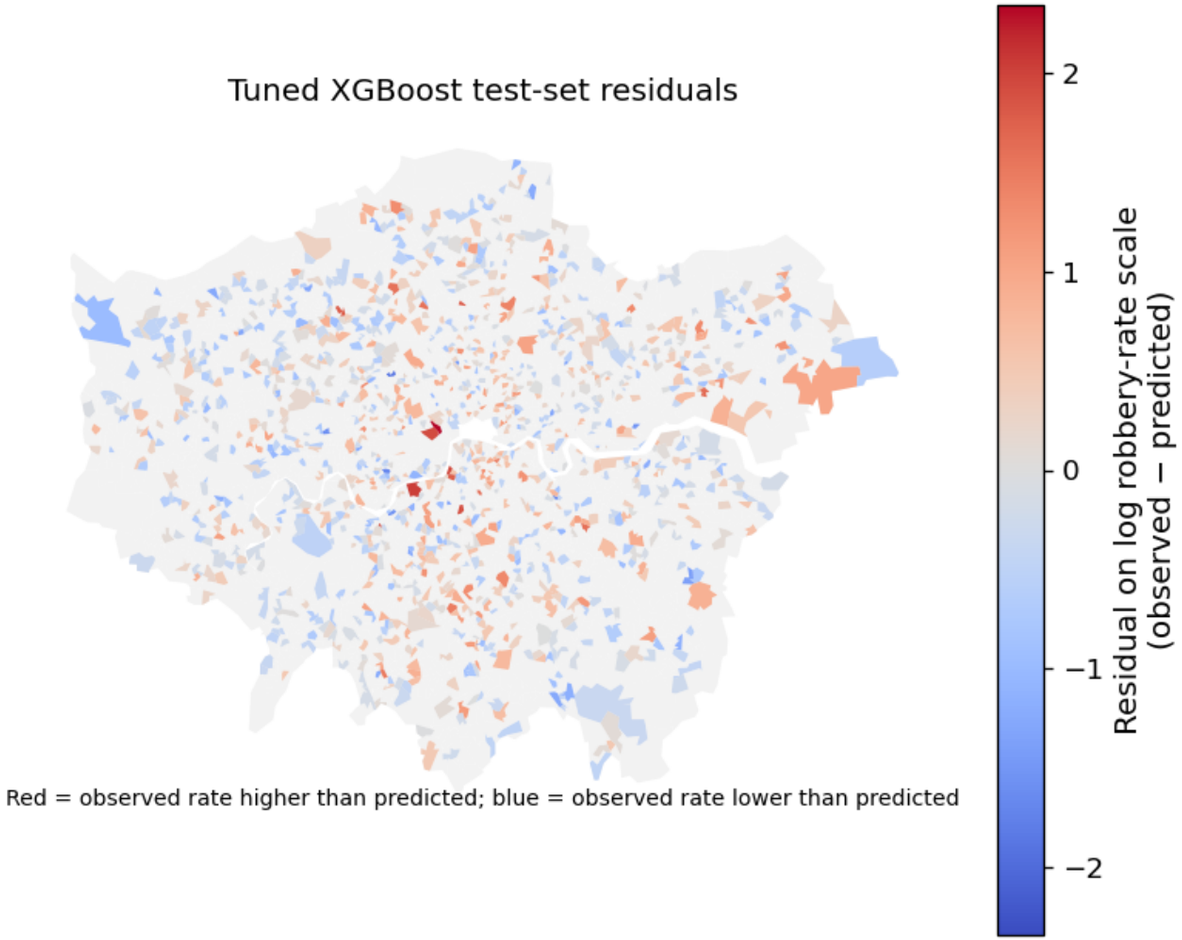
The response rises with PTAL AI. Station distance matters mainly near stations, with the response flattening farther away.

INTERPRETATION LIMIT

Partial dependence describes this fitted model, not a causal effect. Sparse value ranges and correlated urban activity require caution.

6 | Spatial diagnostics and validation

Residual geography and borough-grouped validation assess local prediction errors and spatial sensitivity.



BOROUGH-GROUPED CROSS-VALIDATION

Five-fold borough-grouped cross-validation was conducted within the training data, keeping each borough together. Baseline and extended XGBoost used the same fixed settings.

Mean grouped-CV score	Baseline	Extended
R^2	0.135	0.421
RMSE	0.716	0.584
MAE	0.562	0.467

Transport-related improvement remained under this geographical split. Grouping by borough does not remove all spatial dependence, including similarities across borough borders.

RMSE/MAE are on the log1p rate scale. The grouped R^2 0.421 is not the random held-out R^2 0.479 of the tuned model.

Tuned model, random held-out test areas only. Red: observed > predicted; blue: observed < predicted. Pale background includes areas not evaluated in this test map.

7 | Conclusions and implications

Transport indicators improved prediction across all model families, subject to exposure-measurement and spatial-resolution limitations.

FINDINGS

- Transport variables improved prediction beyond the selected deprivation, density and age baseline.
- PTAL AI was the strongest predictor in the final model; proximity alone carried less information.
- The improvement remained under the borough-grouped sensitivity check.

LIMITS

- Resident population does not capture commuters, visitors or commercial activity.
- LSOA averages and centroid distances simplify real journeys and station surroundings.
- Predictive associations and grouped validation do not establish causality or eliminate spatial dependence.

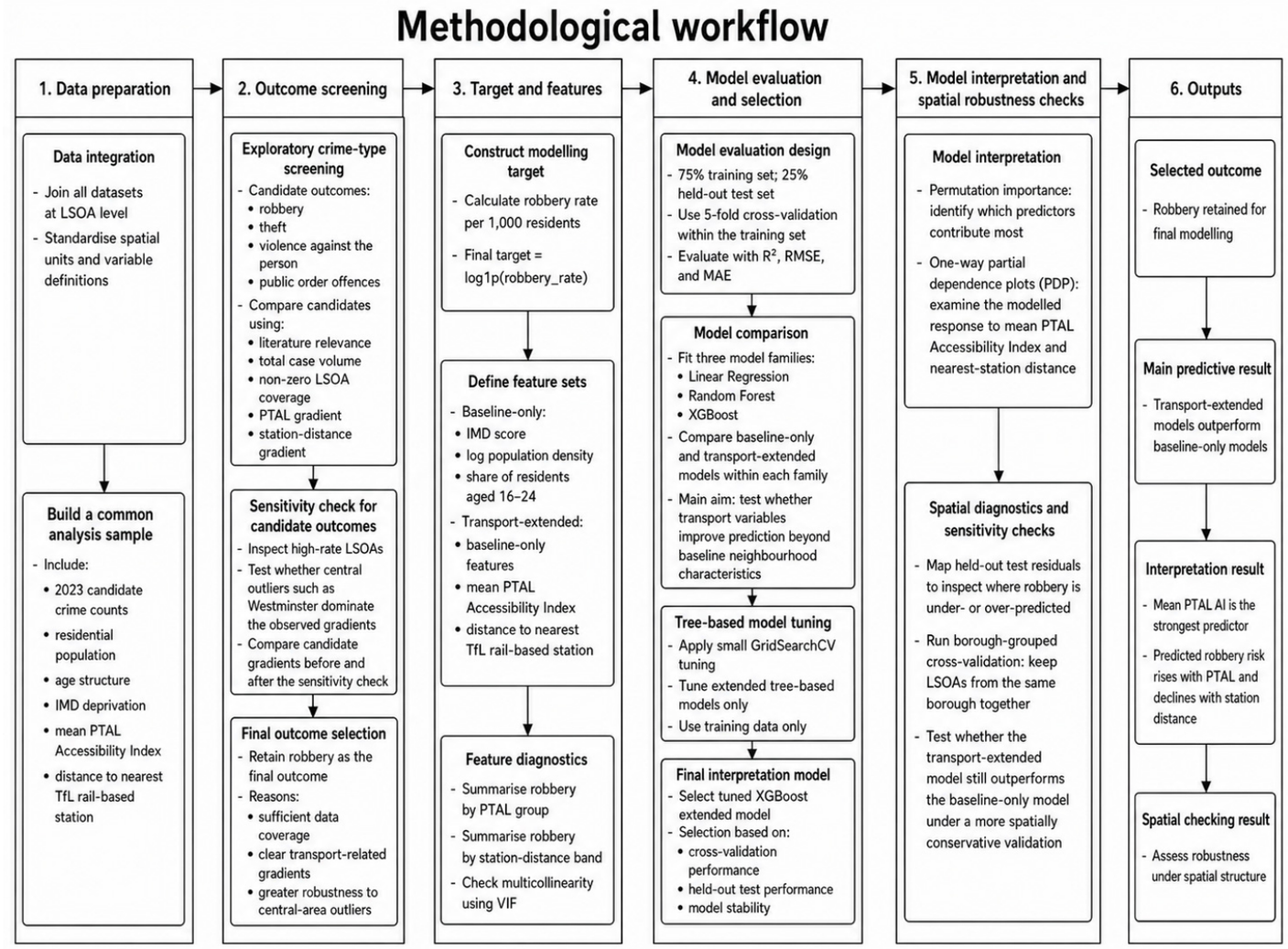
IMPLICATIONS FOR ANALYSIS

- Area comparisons require consistent definitions, geographical units and data coverage.
- Measures and validation designs should reflect the intended geographical application.
- Reporting should distinguish observed patterns, predictive performance and limitations.

Further analysis should examine activity-population and land-use measures, network-based journey accessibility, and spatial and temporal transfer.

Appendix A | Detailed methodological workflow

Data preparation, outcome selection, model evaluation, interpretation and spatial validation.



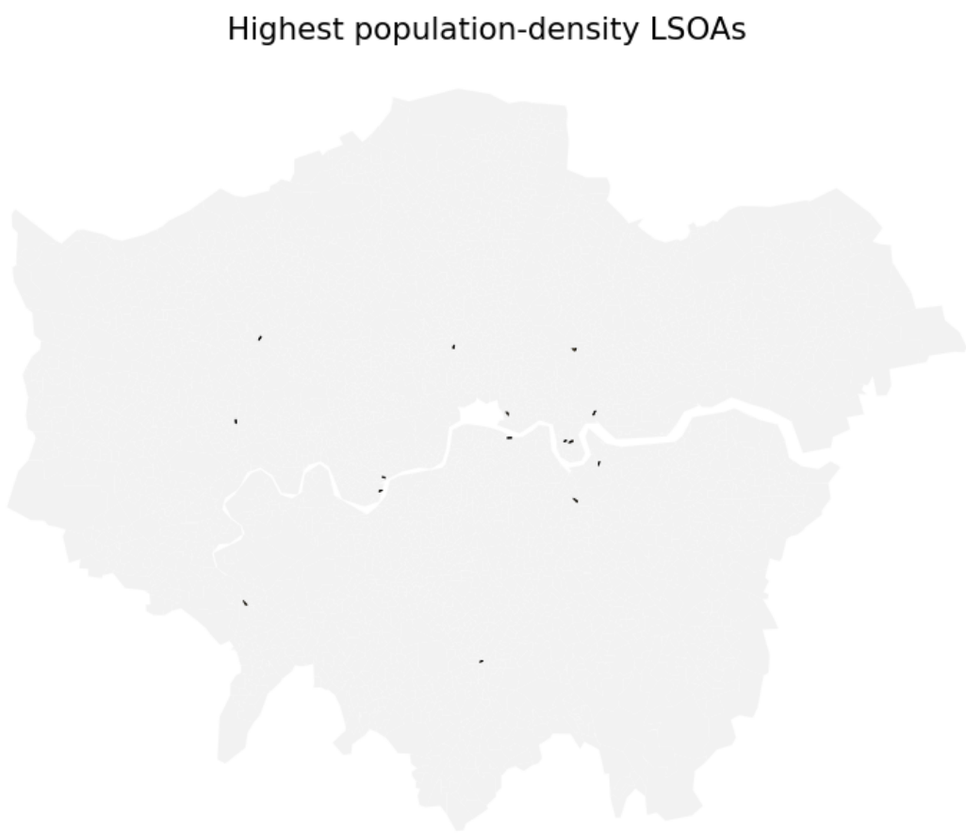
SIX STAGES

- 1. Integrate and check data.
- 2. Screen candidate outcomes.
- 3. Define target and features.
- 4. Compare, tune and select.
- 5. Interpret and check spatial robustness.
- 6. Report findings and limits.

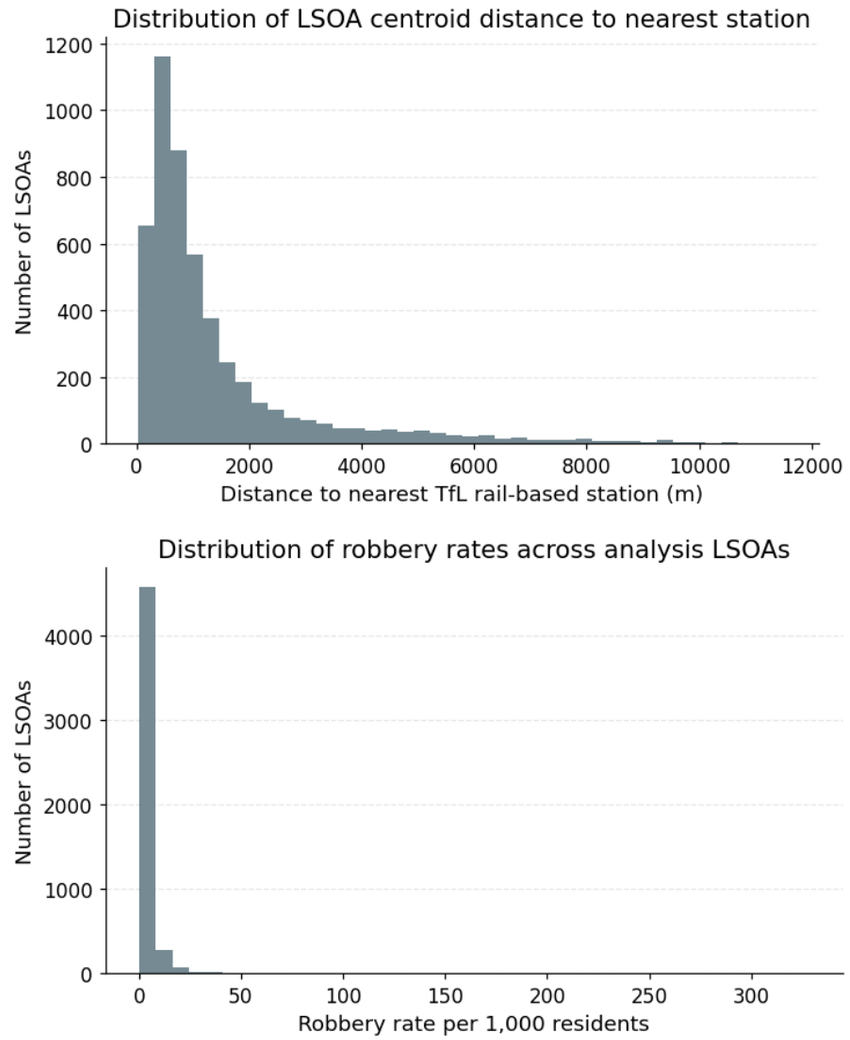
Scope: Westminster exclusion addresses one influential central area; grouped CV tests spatial sensitivity. Exposure limitations and spatial dependence remain relevant.

Appendix B | Spatial and distribution diagnostics

Distributional checks informed feature transformation and examination of population-density extremes.



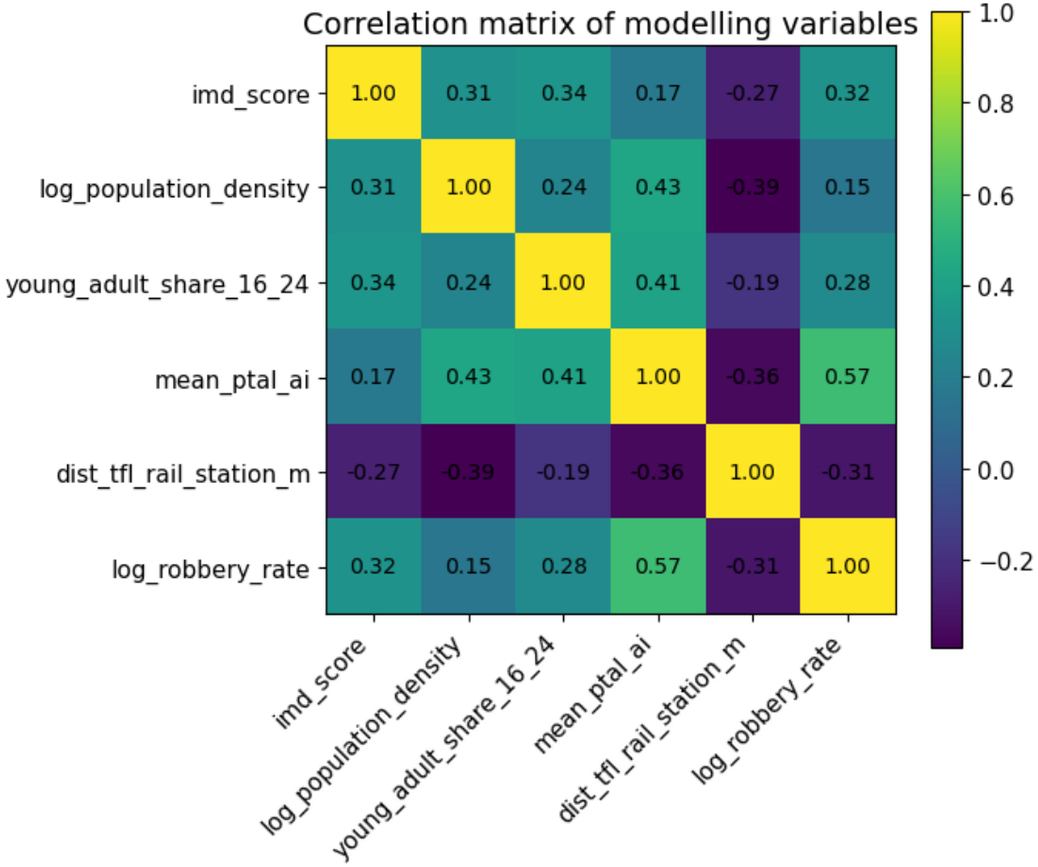
The 15 highest population-density LSOAs are highlighted. Very small polygon areas can generate extreme density values; population and boundary-area consistency are important to interpretation.



Top: centroid-to-station distance, metres. Bottom: raw robbery rate per 1,000 residents. The right-skewed target motivated log1p transformation.

Appendix C | Feature diagnostics and model settings

Three evaluation configurations: fixed comparison, tuned final model and borough-grouped validation.



Predictor VIFs ranged from 1.24 to 1.47, indicating no severe multicollinearity under the selected specification. Confounding remains possible.

Evaluation / configuration	R^2
Fixed XGBoost, random held-out test: baseline / extended	0.199 / 0.482
Tuned extended XGBoost, random held-out test	0.479
Fixed XGBoost, grouped CV mean: baseline / extended	0.135 / 0.421

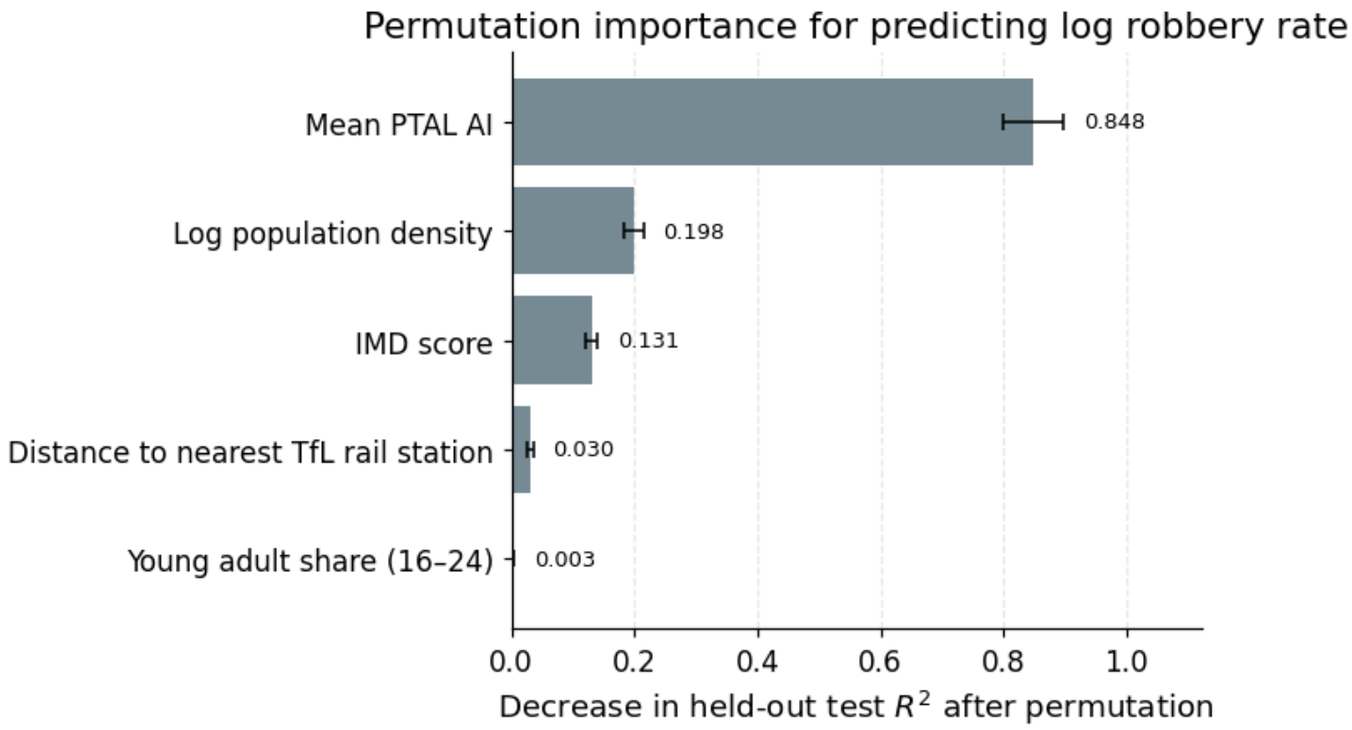
FINAL TUNED XGBOOST

500 trees; maximum depth 2; learning rate 0.03; subsample 0.8; column subsample 0.8; random seed 42. Grid search used training data only.

Tuned-model test RMSE: 0.597; MAE: 0.477, both on the log1p rate scale. These are not errors in offences per 1,000 residents.

Appendix D | Permutation importance

Predictive importance in the final tuned XGBoost model, evaluated on the held-out test set.



PREDICTOR CONTRIBUTIONS

- Mean PTAL AI contributed most to this fitted model; density and deprivation also contributed.
- The nearest-station distance contributed less than the broader PTAL accessibility measure.
- The values are changes in R^2, not percentages of crime explained or estimates of causal effects.
- Interpretation depends on the included features and their relationships.

Bars: mean decrease in test R^2 after permuting one predictor. Error bars: standard deviation over 30 repeats.

Appendix E | Data sources and references

Hanyun Hu | MSc Urban Spatial Science, UCL | 4 November 2026

DATA SOURCES

- Metropolitan Police Service: LSOA-level geographic crime counts, aggregated to 2023.
- Small-area population estimates: 2023 residential population and age structure.
- English Indices of Deprivation: IMD score.
- TfL GIS Open Data: public transport accessibility and rail-based station locations.
- LSOA boundaries: common identifiers and EPSG:27700 geometry for area and distance calculations.

SELECTED REFERENCES

Ceccato, Gaudelet & Graf (2022). Crime and safety in transit environments.
doi:10.1007/s12469-021-00265-1

Irvin-Erickson & La Vigne (2015). Crime at Washington, DC Metro Rail stations.
doi:10.1186/s40163-015-0026-5

Kirchmaier, Langella & Manning (2024). Commuting for Crime.
doi:10.1093/ej/uead089

Li & Kim (2023). Subway Station and Neighborhood Crime.
doi:10.1177/00111287221114803

Liu, Lan, Eck & Kang (2020). Bus stop relocation and street robbery.
doi:10.1016/j.compenvurbsys.2019.101455

Sutherland, Brunton-Smith & Jackson (2013). Collective Efficacy, Deprivation and Violence in London.
doi:10.1093/bjc/azt050